Universe Unfolding

Safety Without Exceptions

Exact Causal Seeds, Observable Memory, and Quantum Experimental Interfaces

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Preserve every admissible choice that preserves everyone.

Abstract

We separate robust safety from probabilistic simulation. Safety is required for *every admissible disturbance*, including those assigned probability zero; it is not a risk budget. We characterize the greatest controlled invariant set, prove indefinite safety under every retained action, and give a terminating algorithm for finite state spaces. For finite experimental trees we prove that exact simulation of all robustly safe policies has precisely the seed cost of a pruned stochastic tree. A reconstruction principle yields exact costs for sum-and-moment memory and sequential Fourier-sector observations. Quantum instruments provide a finite controlled interface incorporating measurement back-action; the two-path interference formula is derived explicitly. All internal propositions have proofs. The sharp additive causal Shannon upper bound is a separately identified external dependency, not proved in this paper. No claim of historical novelty or derivation of cosmology is made.

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1 Scope and the distinction that matters

Three objects must be kept separate: a state or seed, the laws it realizes, and the interface through which it is observed. Equal observable laws need not mean equal hidden states. A zero readout need not mean zero information in a different readout. Conversely, hidden structure cannot be inferred merely because an output is zero.

Our main results are finite probabilistic statements. Signal labels may belong to an infinite-dimensional function space, but every random signal considered below has finite support. Thus its Shannon entropy is discrete entropy, not differential entropy, signal energy, thermodynamic entropy, or description length of an arbitrary real number. Exact arithmetic and computation time are separate resources.

2 A single seed for an adaptive interface

Let T be a finite tree of complete histories. At a nonterminal history h, an experimenter chooses $a \in A(h)$ from a finite nonempty set and observes $y \in Y(h,a)$ from a finite nonempty set, with probability $p(y \mid h,a)$. These probabilities are nonnegative and sum to one. A deterministic policy π selects actions using only the observed history. It does not inspect the hidden seed.

Definition 2.1 (Exact realization)**.** *A finite seed $S \sim \mu$ and a deterministic decoder f realize a class Π of policies if*

$$A_t = \pi(H_{t-1}), \quad Y_t = f(S, H_{t-1}, A_t), \quad H_t = H_{t-1} \,\|\, (A_t, Y_t), \quad H_0 = \varnothing,$$

induce the prescribed transcript law for every $\pi \in \Pi$. The law μ and the decoder f are fixed independently of the selected policy. Write

$$C_\Pi(T) = \inf_{(\mu,f)\ \text{exact for}\ \Pi} \mathrm{H}(S), \qquad C(T) = C_{\text{all deterministic policies}}(T).$$

The shared decoder enforces agreement at shared histories. The quantifiers are $\exists(\mu,f)\ \forall\pi$, not a separate simulator chosen for each policy. A finite realization always exists: pre-sample independently one output for every history–action pair. Along any trajectory, the next table entry is fresh relative to the entries determining the past. This reproduces the conditional laws by induction.

Information envelope. For each policy let X^π be its complete terminal transcript, and put $I_\pi = -\log_2 \Pr(X^\pi)$ on its positive-probability values. All logarithms and entropies below use base two. Define

$$G_T(u) = \min_\pi \Pr(I_\pi \le u), \qquad L(T) = \int_0^\infty (1 - G_T(u))\, du.$$

There are finitely many deterministic policies.

Proposition 2.2 (A self-contained envelope lower bound)**.** *For every finite controlled tree,*

$$L(T) \le C(T) \le \sum_{(h,a)} \mathrm{H}\big(p(\cdot \mid h,a)\big).$$

Proof. Fix a simulator and policy. Its transcript is a deterministic function $g_\pi(S)$. For every seed atom s of positive mass,

$$\Pr\{X^\pi = g_\pi(s)\} \ge \Pr\{S = s\}.$$

Consequently $I_\pi(g_\pi(S)) \le I_S(S) := -\log_2 \Pr(S)$. Thus $\Pr(I_S \le u) \le \Pr(I_\pi \le u)$ for every policy, so $\Pr(I_S \le u) \le G_T(u)$. Integrating the tails of these nonnegative finite-valued random variables gives $\mathrm{H}(S) \ge L(T)$. Taking the infimum proves the lower bound. For the upper bound, independently sample the finite table of outputs for all history–action pairs. Its entropy is the displayed sum. The table construction described above is exact for every policy. □

External sharp upper bound: a precise dependency. The companion project [1] supplies the proposed sharper estimate

$$C(T) \le L(T) + c_\star, \qquad c_\star = \frac{1 + \log_2 e}{2}, \tag{1}$$

for its stated applicable class of trees. Every use of (1) in this manuscript is conditional on that external theorem applying to the particular tree in question. We do not give its proof or certify its current repository implementation here. Without this imported upper bound, Proposition 2.2 and all the exact identities and safety results below remain self-contained. Calling the entire sharp sandwich proved here would be incorrect.

3 Exact reconstruction and observational compression

For a finite-valued random variable X and deterministic q, the chain rule gives

$$\mathrm{H}(X) = \mathrm{H}(q(X)) + \mathrm{H}(X \mid q(X)). \tag{2}$$

Hence q preserves entropy precisely when X is recoverable from $q(X)$ almost surely. The decoder can be defined arbitrarily on values outside the support. A nonzero output alone does not guarantee recoverability.

Theorem 3.1 (Reconstructing-policy principle)**.** *Suppose a finite-support random variable Z drives a deterministic causal interface exact for all $\pi \in \Pi$. If one policy π_0 produces a transcript X from which Z is almost surely recoverable, then*

$$\boxed{C_\Pi = \mathrm{H}(Z).}$$

The minimum is attained by the seed $S = Z$.

Proof. The given interface with seed Z gives the upper bound. Since X is a function of Z and conversely Z is recoverable from X, their entropies agree. Under π_0, every competing exact simulator produces a transcript X' with the same law as X, deterministically from its seed U. Thus $\mathrm{H}(U) \ge \mathrm{H}(X') = \mathrm{H}(X) = \mathrm{H}(Z)$. □

Proposition 3.2 (Restriction and recoding)**.** *If $\Pi_1 \subseteq \Pi_2$, then $C_{\Pi_1} \le C_{\Pi_2}$. Invertible relabelings of actions and outputs that respect histories preserve the optimal seed cost.*

Proof. An exact simulator for the larger class also serves the smaller one. For an invertible relabeling, translate policies and decoder outputs in each direction. This gives simulators of identical seed entropy in both directions. □

An unused independent seed component changes no observable law, while increasing seed entropy. Therefore the hidden realization is generally not identifiable; the optimum is an operational resource quantity.

4 Memory that survives cancellation

Fix real addresses $x_1, \dots, x_n$ and complex deposits $Z = (Z_1, \dots, Z_n)$ with an arbitrary finite-support joint law. Define

$$S_0 = M_0 = 0, \qquad S_t = S_{t-1} + Z_t, \qquad M_t = M_{t-1} + x_t Z_t.$$

At time t the observer may request S_t or M_t. Observation is passive: it does not change the deposits or their law.

Corollary 4.1 (Observable writer)**.** *The optimal common seed entropy is*

$$C_n = \mathrm{H}(Z_1, \dots, Z_n) = \sum_{t=1}^{n} \mathrm{H}(Z_t \mid Z_1, \dots, Z_{t-1}).$$

Proof. Always requesting the sum reconstructs $Z_t = S_t - S_{t-1}$. Apply Theorem 3.1 and the chain rule. □

The update algebra is the writer used in the supplied ONE manuscript [7]; the probability law and observation interface are additional inputs. This use of the writer is not an endorsement of the manuscript's cosmological claims.

For two deposits $Z_1 = W$, $Z_2 = -W$ and $x_1 \neq x_2$,

$$S_2 = 0, \qquad M_2 = (x_1 - x_2)W.$$

With terminal-only observation, the sum interface has cost zero, whereas the interface offering either sum or moment has cost $\mathrm{H}(W)$. A constant seed suffices for the first; seed W suffices for the second and its injective moment readout gives the matching lower bound. This is not true of the sum interface if the earlier sum $S_1 = W$ is also observable.

5 Fourier sectors as an exact observation interface

Use the Fourier convention

$$(\mathcal{F}f)(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi}\, dx.$$

The integral is first defined on Schwartz functions; Plancherel's theorem extends it unitarily to $L^2(\mathbb{R})$. Let h_n be the normalized Hermite basis in this convention, so $h_0(x) = 2^{1/4} e^{-\pi x^2}$ and

$$\mathcal{F} h_n = (-i)^n h_n, \qquad \mathcal{F}^4 = I.$$

These are classical facts, not new consequences of the causal seed theorem. Euler's identity alone is not a proof of them.

For $\Lambda = \{1, -i, -1, i\}$ define

$$P_\lambda = \frac{1}{4} \sum_{j=0}^{3} \lambda^{-j} \mathcal{F}^j.$$

Applying this polynomial to h_n gives h_n if $(-i)^n = \lambda$ and zero otherwise, by the finite geometric-sum identity. Completeness of the Hermite basis consequently yields

$$P_\lambda P_\nu = \mathbf{1}_{\lambda=\nu} P_\lambda, \quad P_\lambda^* = P_\lambda, \quad \sum_{\lambda \in \Lambda} P_\lambda = I.$$

In particular, P_1 is a projection onto the Fourier-fixed subspace, not a universal lossless representation. For example $P_1 h_1 = 0$.

Theorem 5.1 (Sequential Fourier-sector reconstruction)**.** *Let Z be an $L^2(\mathbb{R})$-valued random signal with finite support. In a four-round passive interface, each action requests one of the outputs $P_\lambda Z$, and repeated requests are allowed. The optimal seed entropy for all adaptive policies is exactly* $\mathrm{H}(Z)$*. If only P_1 is available, the optimal seed entropy is* $\mathrm{H}(P_1 Z)$*. The difference is*

$$\boxed{\mathrm{H}(Z) - \mathrm{H}(P_1 Z) = \mathrm{H}(Z \mid P_1 Z).}$$

Proof. The fixed policy requesting each of the four sectors recovers $Z = \sum_\lambda P_\lambda Z$. The seed Z supplies every adaptive sequence of requests. Theorem 3.1 proves the first claim. If only P_1 is available, the first output has entropy $\mathrm{H}(P_1 Z)$ and all repeated outputs equal it. Taking this output itself as seed proves the second claim. Equation (2) gives the difference. □

Corollary 5.2 (A fixed readout with arbitrarily large accessible entropy)**.** *Let W be a finite real random variable and $Z = h_0 + W h_1$. Then*

$$P_1 Z = h_0, \qquad P_{-i} Z = W h_1.$$

The fixed-sector interface costs zero; the interface allowing either of these two sectors costs $\mathrm{H}(W)$*, even in a single round.*

Proof. The first readout is constant; the second uniquely identifies W because $h_1 \neq 0$. Seed W generates both readouts, and the second gives the lower bound. □

Uniform W on m values gives a gap of $\log_2 m$. The hidden information resides in Z, not in the constant output h_0.

Why sequential access is essential. Joint reconstructability of a list of queries is not sufficient if only one query can ever be asked. Let U, V be independent fair bits and $Z = U h_0 + V h_1$. One round allowing either sector query costs only one bit: a single fair seed B can answer $B h_0$ or $B h_1$ as requested. The separate marginals are correct, though this seed does not reproduce their joint independence. Two rounds allowing both queries require two bits, because the fixed policy requesting both reconstructs (U, V). This demonstrates why the experimental interface, including its horizon, belongs in the theorem.

6 Safety without exceptions: the infinite-time invariant

Let X be a set of states and $K \subseteq X$ the required safe set. At state x the controller may choose $a \in A(x)$; nature then chooses $w \in W(x, a)$, where disturbance sets are nonempty. The update is

$$x_{t+1} = F(x_t, a_t, w_t).$$

The controller observes states and its history and chooses before seeing w_t. No probability distribution is needed in this section. We work with ordinary set-theoretic choice when selecting from arbitrary nonempty sets. For finitely many participants $i \in I$, one may specify $K = \bigcap_i \{x : d_i(x) \geq 0\}$ for chosen protection thresholds d_i. These thresholds are model specifications, not numerical values of people.

For $D \subseteq X$ define the controlled predecessor

$$\mathsf{P}(D) = \{x \in K : \exists a \in A(x)\ \forall w \in W(x, a),\ F(x, a, w) \in D\}.$$

A set D is *controlled invariant* if $D \subseteq \mathsf{P}(D)$. Define

$$V^\star = \bigcup \{D \subseteq X : D \subseteq \mathsf{P}(D)\}, \qquad A^\star(x) = \{a \in A(x) : \forall w \in W(x, a),\ F(x, a, w) \in V^\star\}.$$

The union includes the empty set and need not be nonempty.

Theorem 6.1 (Greatest invariant and indefinite protection)**.** *The set $V^\star$ is the greatest fixed point of P. For every $x \in V^\star$, $A^\star(x)$ is nonempty. Starting at $x_0 \in V^\star$, every history-dependent strategy choosing $a_t \in A^\star(x_t)$ satisfies*

$$\boxed{x_t \in V^\star \subseteq K \quad \text{for every } t \geq 0 \text{ and every admissible disturbance sequence.}}$$

Moreover, $V^\star$ is exactly the set of states from which an indefinitely safe strategy exists.

Proof. The map P is monotone. If $x \in V^\star$, it belongs to an invariant set D and has an action with all successors in $D \subseteq V^\star$. Thus $V^\star \subseteq \mathsf{P}(V^\star) \subseteq K$. Monotonicity gives $\mathsf{P}(V^\star) \subseteq \mathsf{P}(\mathsf{P}(V^\star))$. Therefore $\mathsf{P}(V^\star)$ is itself an invariant set and, by the definition of the union, is contained in $V^\star$. This proves equality and maximality. The equality supplies nonempty action sets. Induction proves the displayed claim: the definition of each selected action puts every successor back in $V^\star$. A selector of those actions supplies an indefinitely safe strategy.

Conversely, let W_∞ be the set of initial states admitting some safe strategy. At any such state the first action of a witnessing strategy has all successors in W_∞: after each possible first disturbance, its remaining strategy is a safe continuation from that successor. Hence $W_\infty \subseteq \mathsf{P}(W_\infty)$ and $W_\infty \subseteq V^\star$. The previous paragraph gives the reverse inclusion. □

Proposition 6.2 (Maximal retained freedom)**.** *Every indefinitely safe strategy chooses an action in $A^\star(x)$ at every history reachable under admissible disturbances. Conversely, any sequence of choices from these sets is indefinitely safe when started in $V^\star$.*

Proof. At a reachable history of a safe strategy, each successor has the residual safe strategy as a witness to membership in $V^\star$, by Theorem 6.1. Thus the current action belongs to $A^\star(x)$. The converse is that theorem's induction argument. Unreachable histories impose no condition on the outcome of that particular strategy. □

Proposition 6.3 (Finite-state construction)**.** *When X is finite, start with $D_0 = K$ and iterate $D_{n+1} = \mathsf{P}(D_n)$. The sequence decreases and reaches $V^\star$ after at most $|K|$ strict removals. With finite explicitly listed actions and disturbances, this is an executable finite algorithm.*

Proof. $D_1 \subseteq D_0$, and monotonicity propagates the inclusion. Each strict step removes a state; at a stabilized step $D = \mathsf{P}(D)$. Every invariant set is contained in all D_n, by induction, so $V^\star \subseteq D$. Conversely, a fixed point is invariant and hence contained in $V^\star$. □

For infinite state spaces the finite termination statement does not apply; we make no claim that a countable descending iteration always suffices.

Why the quantifiers cannot be reversed. Take $X = \{s, b\}$, $K = \{s\}$, actions and disturbances $\{0, 1\}$. At s put $F(s, a, w) = s$ if $a = w$ and b otherwise; make b absorbing. For each disturbance there exists a matching safe action, but no action protects against both disturbances. Thus

$$(\forall w\ \exists a \text{ safe}) \not\Rightarrow (\exists a\ \forall w \text{ safe}).$$

Here $V^\star = \varnothing$. A safety requirement alone does not create a controller capable of satisfying it.

7 Exact seed cost under robust safety

We now relate the deterministic guarantee to the finite probabilistic tree. For every history–action pair specify a nonempty set $E(h, a) \subseteq Y(h, a)$ of *admissible* outcomes and require

$$\operatorname{supp} p(\cdot \mid h, a) \subseteq E(h, a).$$

Admissible outcomes may have probability zero. Safety quantifies over E; simulation reproduces p. These are different requirements. Let B be an extension-closed set of forbidden histories. A policy is robustly safe if every path following it through admissible outcomes avoids B.

Define a Boolean winning predicate r by backward recursion:

$$r(h) = \begin{cases} 0, & h \in B, \\ 1, & h \notin B \text{ is terminal}, \\ 1, & h \notin B \text{ is nonterminal and } \exists a\ \forall y \in E(h,a) : r(hay) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

At a winning history retain

$$A_R(h) = \{a \in A(h) : \forall y \in E(h,a),\ r(hay) = 1\}.$$

If $r(\varnothing) = 1$, let T_R be the stochastic tree reached by retained actions and positive-probability outcomes. Its probabilities need no renormalization: removed zero-probability outcomes have total mass zero. Let Π_R be the robustly safe policies on the full tree.

Proposition 7.1 (Winning histories)**.** *A history is winning precisely when a robustly safe continuation from it exists. Every retained action has a winning continuation after every admissible outcome, including zero-probability ones.*

Proof. Backward induction. At forbidden and terminal histories the claim follows from the definition. At a nonterminal safe history a winning action followed by winning child continuations yields a safe policy. Conversely, the first action of any safe policy must have a safe continuation at every admissible child; induction puts all those children in the winning set. □

Theorem 7.2 (Robust pruning preserves the exact optimum)**.** *If the root is winning, then*

$$\boxed{C_{\Pi_R}(T) = C(T_R).}$$

Proof. First, every policy σ on T_R extends to a robustly safe policy $\widehat{\sigma}$ on T. Use its chosen actions on T_R. At winning histories outside T_R, choose any retained action. At losing histories choose arbitrary actions. Starting at the winning root, induction along every admissible path keeps the run at winning histories, so losing histories are unreachable. The prescribed stochastic laws of σ and $\widehat{\sigma}$ agree, because positive-probability paths cannot leave T_R.

Take a finite-seed simulator for Π_R and discard seed atoms of zero mass. Restrict its decoder to T_R. If at a seed–history–action triple it returns an output removed from T_R, replace that output by a positive-probability output, which exists. These replacements change no run of any policy on T_R. Indeed, if a first replacement occurred on such a run for a positive seed atom, the corresponding extended safe policy would have assigned positive probability to a transcript having target probability zero, contradicting exactness. Thus the restricted decoder is exact, with the same seed entropy. This proves $C(T_R) \le C_{\Pi_R}(T)$.

For the reverse inequality take an exact simulator on T_R and extend its decoder arbitrarily to other interfaces in T. Fix $\pi \in \Pi_R$. At every target history of positive probability reached by π, all admissible children have safe residual continuations. The preceding proposition therefore puts its action in $A_R(h)$. Define a policy σ on T_R to agree with π at these histories, choosing any retained action elsewhere. Induction on depth shows their target transcript laws coincide.

Run the restricted and extended decoders with the same seed. On a positive-mass seed atom the restricted run cannot reach a target-zero history: otherwise exactness would give that history positive probability. At all remaining histories σ and π agree, and both decoders use the same rule. Induction therefore makes the two runs identical. The extension is exact for this π, and the argument applies to every $\pi \in \Pi_R$ without changing the decoder or seed. Taking infima proves the other inequality. □

What zero probability does and does not permit. Consider an action with a safe output of probability one and an admissible forbidden output of probability zero. It is almost surely safe but is not robustly safe. Our recursion rejects it. If admissible outcomes are defined to be exactly the positive-probability outcomes, the two safety notions agree for a finite tree: every admissible finite path then has positive probability. Thus the disturbance specification must be explicit.

Corollary 7.3 (Self-contained bounds and the external improvement)**.** *For a winning root,*

$$L(T_R) \le C_{\Pi_R}(T) \le \sum_{(h,a)\in T_R} \mathrm{H}\big(p(\cdot \mid h,a)\big).$$

If the companion sharp bound applies to T_R, then additionally

$$L(T_R) \le C_{\Pi_R}(T) \le L(T_R) + c_\star.$$

Proof. Substitute Theorem 7.2 into Proposition 2.2. The final upper bound is precisely the external implication (1), applied to T_R. □

Connecting finite experiments to indefinite protection. Finite-horizon viability alone does not guarantee indefinite safety: a history may survive the last observed step and have no safe next action. To retain the infinite-time guarantee, construct the depth-n interface from a starting state in $V^\star$ using only actions in $A^\star(x)$ at each reached state. Assume the state is observed, the resulting action and outcome sets are finite, and every stochastic transition is an admissible transition of the controlled system. Denote this finite interface by $T_n^\star$.

Corollary 7.4 (Finite prefixes of indefinitely safe strategies)**.** *The minimum seed entropy for realizing the length-n transcript laws of all indefinitely robustly safe strategies equals $C(T_n^\star)$. Every policy on $T_n^\star$ admits an indefinitely safe extension.*

Proof. Extend a finite policy through admissible zero-probability branches using arbitrary actions in $A^\star$, and after depth n use any selector of $A^\star$. Theorem 6.1 proves indefinite safety. Conversely, the maximal-retained-freedom proposition puts every indefinitely safe strategy's actions in $A^\star$ at every admissibly reachable history. Thus its stochastic prefix law is represented on $T_n^\star$. The restriction and extension of common decoders, including unused seed–history pairs, are exactly the two constructions in the proof of Theorem 7.2. They preserve seed entropy in both directions, proving equality. □

This statement makes no claim of a finite entropy bound uniform in n.

No infinite-entropy assertion follows from indefinite deterministic safety. For example, a finite controller may safely output a fresh fair bit at each step. The first n outputs require seed entropy at least n, since their joint entropy is n. No single discrete seed of finite entropy realizes all these prefixes, even though the safety invariant holds forever.

8 Invariants and quantum experimental interfaces

For a smooth controlled system $\dot{x} = F_a(x)$, a differentiable quantity U is conserved along every fixed-action trajectory when $dU_x(F_a(x)) = 0$ for every action and state. An intervention Θ_a preserves that quantity only when $U \circ \Theta_a = U$ is separately established. Commuting with the flow is insufficient: scaling the harmonic oscillator state by two commutes with its flow but multiplies $q^2 + p^2$ by four. The conservation statement follows from the chain rule: $\frac{d}{dt}U(x(t)) = dU_{x(t)}(F_a(x(t))) = 0$ on the interval of existence. For piecewise fixed actions the same reasoning applies on each interval; state jumps additionally require the intervention identity. Such identities alone do not specify a probability law or a seed entropy.

For quantum experiments, a precise bridge can be specified at the level of instruments. Let ρ_h be the conditional density operator and let $\{\mathcal{I}_{a,y}\}_y$ be completely positive trace-nonincreasing maps whose sum is trace preserving. Then

$$p(y \mid h, a) = \operatorname{tr} \mathcal{I}_{a,y}(\rho_h), \qquad \rho_{hay} = \frac{\mathcal{I}_{a,y}(\rho_h)}{p(y \mid h, a)} \quad \text{when } p(y \mid h, a) > 0.$$

Assume a fixed initial density operator, a finite horizon, finite action and outcome sets, and an instrument specified at each relevant history. A convention at zero-probability histories is sufficient to define a probabilistic tree, but is not sufficient to establish physical robustness there: admissible zero-probability branches need a separately specified continuation model if they are to enter the robust guarantee.

Proposition 8.1 (Quantum instruments produce an exact finite interface)**.** *The displayed conditional probabilities define a finite controlled tree. It has a finite-seed exact realization for every deterministic adaptive policy. If a supplied admissibility and safety specification has a winning root, its robustly safe policies have simulation cost $C(T_R)$.*

Proof. Complete positivity makes each output operator positive, so its trace is nonnegative. Trace preservation of the sum gives $\sum_y p(y \mid h, a) = \operatorname{tr}\rho_h = 1$. For a positive-probability outcome the normalized output is positive with unit trace, so induction defines every positive-probability history. Multiplying the conditional probabilities along a policy's path gives its quantum sequential outcome law. Finiteness gives the tree and the independent table construction gives an exact common seed. The last assertion is Theorem 7.2; the quantum model does not supply its safety premise. □

Two paths: the interference term, explicitly. Let

$$|\Psi\rangle = |\psi_1\rangle \otimes |d_1\rangle + |\psi_2\rangle \otimes |d_2\rangle$$

be normalized, with normalized detector states. The path amplitudes include all weights. Expanding the outer product and tracing the detector yields

$$\rho_{\text{screen}} = \sum_{i,j=1}^{2} |\psi_i\rangle\langle\psi_j|\langle d_j|d_i\rangle,$$

because $\operatorname{tr}(|d_i\rangle\langle d_j|) = \langle d_j|d_i\rangle$. Taking the position diagonal gives

$$\boxed{p(x) = |\psi_1(x)|^2 + |\psi_2(x)|^2 + 2\operatorname{Re}\left[\psi_1(x)\overline{\psi_2(x)}\langle d_2|d_1\rangle\right].}$$

The two off-diagonal terms are complex conjugates, which proves the formula. Identical detector states preserve the cross term; orthogonal detector states make it zero in the unconditioned screen distribution. This conclusion follows from the joint state and partial trace, not from a human reading the detector. For a finite measurable detector partition $(B_y)_y$ covering the outcome space, the recorded probabilities are $p_y = \int_{B_y} p(x)\,dx$. Other measurement choices and their state updates specify other actions. This is the standard quantum account [6, 5].

A fully computed minimal example. Prepare $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and allow a single choice between projective measurement in the Z or X basis. The Z outcome is a fair bit; the X outcome is always $+$. A single fair seed B supplies B for Z and $+$ for X, hence $C \le 1$. The Z transcript has entropy one, so every seed has entropy at least one. Therefore $C = 1$ bit exactly. If all outcomes are declared safe, the robust pruning changes nothing. If outcome 1 of the Z

action is forbidden, that action is rejected, only X remains, and $C(T_R) = 0$. Safety changes the admissible interface; it does not alter the Born probabilities of the rejected experiment.

The defensible direction is

$$\text{physical model and protocol} \longrightarrow p(y \mid h, a) \longrightarrow T \longrightarrow T_R \longrightarrow C(T_R),$$

where constructing T_R additionally requires explicit admissibility and safety data. A displayed Lagrangian alone omits preparation and measurement. A passive writer does not automatically encode measurement back-action. No Lagrangian, Born rule, gravitational dynamics, or consciousness mechanism is derived here. Classical centralized simulation of a temporal interface does not establish spatially local hidden variables or efficient computation.

9 A conditional estimate is not a proof of RH

The supplied Fourier manuscript also proposes a conditional bound of the form

$$\frac{Q(\tau, L) - E_0 L}{L} \geq A\tau^4 - B\tau^3 - D, \qquad A > 0, \quad B, D \geq 0,$$

valid uniformly for $L \geq 1$ and $\tau \geq \tau_0$. Define

$$\tau_\star = \max\left\{\tau_0, 1, \frac{2B}{A}, \left(\frac{4D}{A}\right)^{1/4}\right\}.$$

Then for $\tau \geq \tau_\star$ the right side is strictly positive: it is at least $A\tau^4/2 - D$, which is at least $D > 0$ if $D > 0$, and is positive if $D = 0$. This is a valid elementary implication. It does not prove the assumed bridge or quartic coercivity estimates. Fourier-sector decomposition and seed entropy do not supply those estimates. No proof of the Riemann hypothesis is claimed in this synthesis.

10 What is established, and what remains to construct

The operational content of unfolding is a sequence of choices and responses. A common seed realizes their prescribed laws across policies. Safety restricts which policies are admissible before an entropy optimization is performed. The principal results are

$$\begin{aligned}
V^\star &= \mathsf{P}(V^\star), \\
x_0 \in V^\star,\ a_t \in A^\star(x_t) &\implies x_t \in K \quad \text{for every admissible run}, \\
C_{\Pi_R}(T) &= C(T_R), \\
C_{\text{reconstructing interface}} &= \mathrm{H}(Z), \\
C_{\text{four sequential sectors}} - C_{\text{fixed sector}} &= \mathrm{H}(Z \mid P_1 Z).
\end{aligned}$$

The first two statements hold for the stated abstract controlled system. The entropy identities are finite-support, finite-interface results. The existence of a safe initial state is not automatic. Nor is a physical system guaranteed to satisfy the chosen admissibility model.

The next research question has zero tolerance for failure. Given a concrete system, determine $V^\star$, prove that the initial state lies in it, retain every action in $A^\star$, and quantify the exact seed cost of its finite experimental interfaces. Additional actuators can enlarge the viable set; additional admissible disturbances can shrink it. These follow directly from the existential action and universal disturbance quantifiers. The research task is to construct and certify these sets for useful systems, not to substitute an allowable failure probability for the safety constraint.

Limits of identification. Exact observational agreement does not identify the hidden implementation. If two models have the same transcript law under every available policy, then every statistic of such a transcript also has the same law: the probability of a statistic's event is the probability of its preimage. No such statistic can therefore distinguish the models with certainty. A scientific distinction requires an experiment for which their laws differ.

Proof and formalization status. All propositions and theorems introduced here have mathematical proofs. Standard finite entropy identities, Plancherel's theorem and Hermite-basis completeness are background results. The sharp additive upper bound (1) is explicitly external; its proof is not supplied. This document is not a new Lean certificate. A supplied report names `CausalSeed/NobodyLeft.lean` and commit `5ae8404`; that report has not been independently checked in preparing this revision, and does not by itself certify every theorem of this manuscript. No historical priority is claimed. No implication concerning $P = NP$, RH, physical immortality or information as matter follows from these results.

Safety is the constraint. Choice remains within it.

Entropy measures the simulation, not the worth of a person.

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